

Project 1

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1. What each member completed:

- a. Josh: Implemented all necessary methods within MainGUI and implemented the relevant extra credit methods as well. Implemented all necessary helper functions to assist in connecting the MainGUI to the LinkedList and streamlining the usage of the LinkedList. (MainGUI, Extra Credit, Helper Methods)
- b. Jordan: Implemented all necessary methods within the LinkedList class, creating its baseline functionalities as listed. Implemented the entire Main (test) class as required. (LinkedList, Main)
- c. Both: PDF Report

2. The proposed project uses a linked list. Could you explain why a linked list was suggested?

Events are constantly being shifted and changed, so we need a list that can dynamically change size and be modified. LinkedList was recommended specifically so we could insert/remove a record in the middle of the list, unlike ArrayLists, which need to shift down or up to fill in the gaps created by a remove() call.

What will be another alternative solution?

Utilizing an ArrayList would be an alternative solution, as it would allow the list to still dynamically change and be modified after creation, but would also simplify some features of the linked list, like remove or add.

3. Explain your design:

a. How did you design your linked list?

I created a singly-linked list containing a Session data point and a link to the rest of the list. This linked list only contains a reference to the first item within the list and does not contain references to its size or last element, identical to the LinkedList implemented in Lab 2.

b. How many classes did you use?

Just 4 classes: the MainGUI class, the Session Class, the Main(test) class, and the LinkedList class.

c. Did you add any helper functionalities?

Yes.

i. Explain them if you did

The `.add(Session)` function was added to the LinkedList to streamline the process of adding a Session to the list at either the start, end, or somewhere in the middle, using an if-else statement. The `.getLast()` function was added to allow easy access to the last element of the list by

simply iterating through it. The `.getByID(id)` function was added to allow access to the actual Session by id rather than simply getting the string value of said session, and the `.getByMentor(mentor)` to serve a similar function but for mentor. These operate exactly like the `searchBy` methods, but return the session rather than a string.